

# Computing for Analytical Work

**Session 3 — What broke?**

**Block 6 — Data files, formats, size, and recovery**

## **This block in one sentence**

**Data problems often look like code problems. Inspect the input before blaming the code.**

The code is the same today as yesterday. The file is not. Look at the file.

# Agenda

1. Data files as external reality
2. What a CSV really is: delimiters and header rows
3. Look at the file *before* you load it
4. Five ways a CSV lies, and the `read_csv` option for each
5. The four-line sanity check
6. Git thread — the final clean submission
7. AI thread — hypotheses from the model, validation from the data
8. Lab 6 — The data is the bug, then Homework 3

## Data files as external reality

- Your code encodes **assumptions**: this column exists, this is a number, this is a date
- The file was made by someone else, on another machine, with another tool, possibly by hand
- The file does not know about your assumptions and does not care
- Same code, new file, new failure — or worse, new *silent* wrong answer
- Session 1 taught you the filesystem is the source of truth. So is the file's **content**

## What a CSV actually is

```
id,name,city  
1,Anna,Vienna  
2,Bela,Budapest
```

- Plain text; one **row per line**
- A **delimiter** separates the fields — comma by convention, not by law
- The **header row** is the first line, and it is only a header because you say it is
- `pd.read_csv` assumes: comma, header on line 1, **NA** -style spellings for missing

## Look at the file before you load it

```
head -5 data/raw/customers.csv  
wc -l data/raw/customers.csv  
file data/raw/customers.csv
```

- `head -5` — the first five lines: delimiter, header, first values, visible at once
- `wc -l` — how many lines; compare with `df.shape` after loading
- `file` — what kind of file this is; catches a renamed Excel file or a zip pretending to be a CSV
- Thirty seconds, in the terminal, before a single line of Python

## Wrong delimiter: what pandas does with it

```
id;name;city  
1;Anna;Vienna  
2;Bela;Budapest
```

```
df = pd.read_csv("data/raw/customers.csv")  
print(df.shape)           # (2, 1)  
print(df.columns.tolist()) # ['id;name;city']
```

**One column with everything in it.** No error. Just a wrong dataframe.

## The fix is one argument

```
df = pd.read_csv("data/raw/customers.csv", sep=";")  
df = pd.read_csv("data/raw/export.tsv", sep="\t")  
print(df.shape)           # (2, 3)
```

- `sep=` tells pandas what the delimiter really is
- You knew which one to pass because you ran `head -5` first
- One column where you expected many is the signature — learn to recognise it

## Extra header rows, missing header rows

```
Orders export
Generated 2026-03-01
order_id,customer_id,amount,order_date
1001,7,"€1,200",2026-01-05
```

```
df = pd.read_csv("data/raw/orders.csv", skiprows=2) # junk above the header
df = pd.read_csv("data/raw/bare.csv", header=None, names=["id", "name", "city"]) # no header at all
```

- Extra lines above: the "header" becomes `Orders export` and every row is misaligned
- No header: your first data row becomes the column names

## Missing values, and the many spellings of missing

```
df = pd.read_csv("data/raw/orders.csv", na_values=["-", "n/a", "missing", "?"])
df.isna().sum()
```

- pandas recognises a few spellings by default: empty, `NA`, `NaN`, `null`, `n/a`
- It does **not** recognise `-`, `?`, `missing`, `.`, or `999` — those stay as text
- One `-` in a numeric column turns the **whole column** into strings
- `na_values=` declares your file's spellings; then `isna().sum()` shows you where they were

## Numbers stored as text

```
df["amount"].head()           # 0    €1,200    dtype: object
df = pd.read_csv("data/raw/orders.csv", thousands=",")           # handles 1,200
df["amount"] = (df["amount"].str.replace("€", "", regex=False)
               .str.replace(",", "", regex=False))
df["amount"] = pd.to_numeric(df["amount"], errors="coerce")       # bad ones become NaN
```

- `dtype: object` on a column that should be numeric is the tell
- Strip the symbols, then convert; `errors="coerce"` turns leftovers into `NaN` you can count

## Dates as strings

```
df = pd.read_csv("data/raw/orders.csv", parse_dates=["order_date"])
df["order_date"] = pd.to_datetime(df["order_date"], errors="coerce", format="mixed")
df["order_date"].isna().sum()    # how many dates failed to parse?
```

- A date column with `dtype: object` is text, and sorting text dates gives nonsense
- `parse_dates=` for well-behaved files; `to_datetime(errors="coerce")` for messy ones
- Mixed formats ( `2026-01-05` , `05/01/2026` , `5 Jan 2026` ) need `format="mixed"` and a **count of failures**
- `05/01/2026` is ambiguous — day first or month first? The file's author knows; you must ask

## The four-line sanity check

```
df = pd.read_csv("data/raw/sales.csv")
print(df.shape)      # (rows, columns) – is this the size you expected?
print(df.dtypes)      # numbers, not strings? dates, not objects?
df.head()             # what does the data actually look like?
df.isna().sum()       # which columns have gaps, and how many?
```

**Run it on every dataframe, every time — as reflexively as you hit save.**

## What each line catches

- `df.shape` → wrong delimiter (1 column), skipped rows, truncated file
- `df.dtypes` → numbers as text, dates as text, one stray `–` poisoning a column
- `df.head()` → junk header rows, misaligned fields, a currency symbol you did not expect
- `df.isna().sum()` → unrecognised spellings of missing, coercion that silently ate values
- Also look at: `df.columns.tolist()` for names, `df["col"].unique()[:10]` for surprises
- Compare every number to what you expected **before** you loaded the file

## Symptom in the code, cause in the file

| You see                                                        | Look at the file for                                                   |
|----------------------------------------------------------------|------------------------------------------------------------------------|
| <code>KeyError: 'name'</code> after load                       | wrong delimiter, extra header lines, different header spelling         |
| <code>ValueError: could not convert</code>                     | currency symbols, thousands separators, a <code>-</code> for missing   |
| <code>TypeError</code> on <code>+</code> or <code>sum()</code> | a text column that should be numeric                                   |
| A total that is 10× too small                                  | rows dropped by <code>skiprows</code> , or coerced to <code>NaN</code> |
| A date sort that looks random                                  | dates still stored as text                                             |

## Not today: Excel, JSON, encodings, file size

- All four are real and you will meet them this year
- They live on the **Lab 6 reference sheet** — "Other formats and other problems" — in the lab README
- Excel: why opening data in Excel silently changes it
- JSON: what malformed means and how to spot it
- `encoding="latin-1"` when `utf-8` fails on the first byte
- Size: why a file may not open in Excel at all, and what to do instead

## **Git thread — the final clean submission**

## `git status` → `git diff` → commit

```
git status
git diff
git add -A
git commit -m "Load orders with correct delimiter, skiprows, and na_values"
git status
```

- `status` : what is modified, what is new, what is untracked junk
- `diff` : read every changed line — is each one something you meant?
- Commit with a message that says **what** and **why**; `status` again should be clean

## Before submitting, know exactly what changed

- If `git diff` surprises you, you are not done
- Leftover `print()` calls, a hard-coded absolute path, a commented-out block — these are in the diff, so you will see them
- Generated output and `.venv/` should be ignored, not committed; `git status` shows if they are not
- `git log --oneline` should read like a story of the repair
- Homework 3 asks for this evidence explicitly: status, diff, final commit

**AI thread — hypotheses from the model, validation from the data**

## Three good questions to ask

1. Here is the traceback and the first five lines of the file (head -5).  
What are the likely causes of this parsing error?
2. I have a 2 GB CSV I have never seen. How do I inspect it safely  
without opening it in Excel or loading all of it?
3. What does `skiprows=` do in `pd.read_csv`, and when would I use  
`header=` instead?

Each one comes back with hypotheses or explanations — **not** with a verified dataframe.

## Validation comes from inspecting the data

- The model suggests `sep=";"`. Did `df.shape` change from `(500, 1)` to `(500, 6)`?
- The model suggests `na_values=["-"]`. Does `isna().sum()` show a count you can explain?
- The model suggests `errors="coerce"`. How many values became `NaN`, and which ones?
- **Verification standard:** row counts, column names, dtypes, sample records, the output file exists
- Your AI-use note says what you asked and which of these you checked

## **5-minute stand-and-stretch**

Stand up. Leave the laptop. Back in five for Lab 6.

## Lab 6 — The data is the bug

The code in `scripts/load.py` is **mostly reasonable**. The input files are not.

- `customers.csv` — semicolon-delimited
- `orders.csv` — two title lines above the real header
- Amounts like `"€1,200"`
- Dates in **mixed** formats
- Missing values spelled `n/a` and `–`

Start in the terminal, not the editor: `head -5`, `wc -l`, `file`.

## Run it, fix the load, validate, note it

```
uv sync
uv run python scripts/load.py
```

- Load each file with the **right options**: `sep=` , `skiprows=` , `na_values=` , `thousands=` , date parsing
- Validate: row count matches `wc -l` , column names match the header, dtypes are right, `isna().sum()` is explainable
- Produce `output/revenue_by_region.csv` and check it exists; one note per data problem in `DIAGNOSIS.md`
- Write the "what could break next time" note in `NEXT_TIME.md` : three things, and the check that catches each
- `git status` , `git diff` , commit — then, at minute 33, explain one file's problems and your validation to your neighbour, about a minute

## The last ten minutes — explain it to a neighbour

**Speaker, about a minute, pointing at the screen, not reading the note:** symptom → cause → evidence → change → verification.

**Listener asks:**

1. Show me the evidence.
2. Why did it fail — not just where?
3. Next month's export gains a fifth column at the end, and nothing else changes. Which of your four checks fires, and does your loading fix still run?

Swap. Unsure, or you disagree? **Hand up.** Then the answer, for everyone. Before you leave: **DIAGNOSIS.md** to the Lab 6 checkpoint on Moodle.

## Homework 3 assigned

### Debugging, data files, recovery, and final diagnosis note

- Expected time: **7–8 hours**
- Due: **Fri 9 Oct, 23:59** — one day late at –10%; the self-check key posts Sunday 11 Oct
- A repo with several realistic failures: **some are code, some are data**
- Repair it, verify the output, explain every failure — and tell code problems from data problems
- Then a **mock exam** in the real format, with a key. The exam is in **reading week**.

## Homework 3 — deliverables

1. Fixed repo with a **clean end-to-end run**
2. **Verified output** file, plus your **data-loading checks**
3. **Final clean commit** — `git status`, `git diff`, commit evidence
4. One **diagnosis note per failure**, standard template
5. **AI-use note** — what you asked, how you verified, if you used it
6. **Video walkthrough, up to 2 minutes** — run it end to end, explain the failures
7. **End-of-course reflection, ~300 words** — next slide

## End-of-course reflection — four prompts, ~300 words

1. Of your three diagnosis notes (HW1, HW2, HW3), which is your **strongest**, and why? **Quote one sentence** from it.
2. Where in the loop (observe → locate → hypothesise → inspect → change → verify → explain) did you most often **cut a step**? What did it cost you?
3. Which class of failure was **slowest** for you — paths, environments, notebook state, tracebacks, or data? What is your concrete plan for getting faster next semester?
4. How did your **use of AI change** between HW1 and HW3? One thing you **stopped** doing, one thing you **started**.

300 words is a ceiling. 200 specific words beat 300 general ones.

## The final exam

Separate session, closed book, in reading week — about four weeks from now. A mock exam with a key comes first. New scenarios, same skills.

**The best preparation is re-reading your own three diagnosis notes.**